

Adrián Soto Suárez

Technical Leader · PhD in Computer Science

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Santiago, Chile

WORK EXPERIENCE

Mar 2025 – Present

Technical Leader · **The Not Company**

Lead the technical side of a multidisciplinary team that brings together software and ML engineers, alongside product and design, building a software platform that empowers users to create product formulations — combining robust data ingestion and processing pipelines with techniques such as AI-based information extraction and information retrieval. Focus on solution design and software architecture, shaping systems that remain reliable and scalable across cloud environments.

Also manage part of the team, focusing on their personal and professional growth: guiding career development while overseeing the technical direction of the products and ensuring best practices at both the architecture and code level.

Mar 2024 – Mar 2025

R&D Director · **IMFD**

Directed a multidisciplinary team of full-stack software engineers, data engineers, data scientists, and UX/UI specialists, focused on developing data products capable of processing large volumes of information using analytics and artificial intelligence techniques. Among the most notable projects, implemented embedding techniques to enable recommendation systems and employed Retrieval-Augmented Generation (RAG) methodologies to allow clients to interact conversationally with their data.

Oversaw the technical leadership of the team, taking responsibility for software design and architecture to ensure scalability for large datasets. Also guided each team member's career development and led product-level solution design, enabling participation in capital-raising initiatives and securing public project bids.

Jan 2022 – Mar 2024

Tech Lead & Manager, Data Team · **Fintual**

Tech Lead and Manager of the Data Team, whose main responsibility is to make data accessible so that analysts from other areas can create value and make decisions based on data. As a manager, focused on the technical and career growth of the team members. On the technical side, led the design of Fintual's data infrastructure, built around a BigQuery data warehouse.

The stack uses DBT to maintain the views within the data warehouse, Airflow as the orchestrator for the data pipelines, and a Kubernetes cluster to execute jobs with efficient and scalable performance. Additionally, implemented near real-time data ingestion with Google Datastream for part of the data. Responsible for researching these technologies and configuring them in a cloud environment to address the team's needs, while maintaining best practices at the code level for the pipelines, as well as for the views and queries within the data warehouse.

Jul 2020 – Dec 2021

Assistant Professor · Universidad Adolfo Ibáñez

Taught data science courses, including a graduate-level course on the fundamentals of Machine Learning and an undergraduate introduction to data science. Conducted research on efficient query processing and fairness in machine learning.

2016 – 2020, 2023

Lecturer · PUC Chile

Taught data-related courses during the PhD, mainly an undergraduate introduction to data management. Received several teaching awards, including Best Part-Time Professor and Most Inspiring Professor in the Computer Science department.

Apr 2015 – Jan 2016

Software Engineer · Simula UC (now Notus)

Full-stack web development of applications solving complex engineering problems using optimization and simulation techniques.

EDUCATION

2016 – 2021

PhD in Computer Science · PUC Chile

Supervisor: Prof. Juan L. Reutter. Thesis: *Efficient processing of recursive and federated queries in SPARQL*. Research on graph databases and efficient query processing: extending graph query languages with analytics features, integrating graph data with web sources, improving index structures for graph-based storage, and designing novel query algorithms. The work spanned theory and practice, including extending systems such as Apache Jena and deploying them to production environments for real-world testing.

2011 – 2018

Software Engineering degree · Faculty of Engineering, PUC Chile

TEACHING EXPERIENCE

PUC Chile — School of Engineering

- **IIC2440 · Big Data Processing** — Lecturer (Autumn 2023, 2024)
- **IIC2413 · Databases** — Lecturer (Spring 2016–2018, Autumn 2019, 2020)
- **Graph Theory for Big Data** — Postgraduate diploma lecturer (Spring 2019, Winter 2020)
- **Data Journalism** — Organizer and lecturer of a one-week course on data management and programming for journalists (2017, 2018)
- **Web Scraping for Political Science** — Guest lecture in a programming course for political scientists (2019)
- **Java for School Students** — Organizer and teacher (2017, 2018)
- **Teaching Assistant** — During my undergraduate career I was teaching assistant of the following courses:
 - IIC2233 · Advanced Programming (×4)
 - IIC1103 · Introduction to Programming (×3)
 - IIC2413 · Databases (×3)
 - IIC2154 · Capstone Project (×2)
 - IIC3413 · Database Systems Implementation (×2)
 - IIC2143 · Software Engineering
 - IIC2513 · Web Technologies and Applications
 - IIC3242 · Computational Complexity
 - IIC3432 · Advanced Topics in Databases
 - IIC3272 · Cryptocurrencies and Smart Contracts

UAI Chile — Faculty of Engineering and Sciences

- **ING559 · Machine Learning Methods for Data Science** — Lecturer (Spring 2020, 2021)

- **TICS575 · Big Data Techniques** — Lecturer (Autumn 2021)
- **TICS580 · Python & Relational Databases Bootcamp** — Lecturer (Summer 2021)
- **TICS314 · Fundamentals of Data Science** — Lecturer (Spring 2021)
- **Machine Learning** — Postgraduate diploma lecturer (August 2021)

JOURNAL PUBLICATIONS

- D. Arroyuelo, A. Gómez-Brandón, A. Hogan, G. Navarro, J. L. Reutter, J. Rojas-Ledesma, A. Soto — The Ring: Worst-Case Optimal Joins in Graph Databases using (Almost) No Extra Space. *ACM TODS* 49(2), 2024
- M. Mosser, F. Pieressa, J. L. Reutter, A. Soto, D. Vrgoč — Querying APIs with SPARQL. *Information Systems Journal*, 2022
- J. L. Reutter, A. Soto, D. Vrgoč — Recursion in SPARQL. *The Semantic Web Journal*, 2021
- L. Libkin, J. L. Reutter, A. Soto, D. Vrgoč — TriAL: A navigational algebra for RDF triplestores. *ACM TODS* 43(1), 2018

CONFERENCE PUBLICATIONS

- D. Arroyuelo, A. Hogan, G. Navarro, J. L. Reutter, J. Rojas-Ledesma, A. Soto — Worst-case Optimal Graph Joins in Almost No Space. *SIGMOD* 2021
- A. Hogan, J. L. Reutter, A. Soto — In-Database Graph Analytics with Recursive SPARQL. *ISWC* 2020
- A. Hogan, C. Riveros, C. Rojas, A. Soto — A Worst-Case Optimal Join Algorithm for SPARQL. *ISWC* 2019
- M. Mosser, F. Pieressa, J. L. Reutter, A. Soto, D. Vrgoč — Querying APIs with SPARQL: Language and Worst-Case Optimal Algorithms. *ESWC* 2018
- J. L. Reutter, A. Soto, D. Vrgoč — Recursion in SPARQL. *ISWC* 2015

WORKSHOPS & DEMOS

- M. Mosser, F. Pieressa, J. L. Reutter, A. Soto, D. Vrgoč — Querying APIs with SPARQL. *AMW* 2019
- J. Castro, A. Soto — A Comparison between Cypher and Conjunctive Queries. *AMW* 2017
- M. Junemann, J. L. Reutter, A. Soto, D. Vrgoč — Incorporating API Data into SPARQL Query Answers. *ISWC Demos* 2016